Agent communication over MoQ (Dapeng)

Agent Protocol over MoQ

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|     Agent to Agent Communication - Application Logic            |
|  (Agent Reasoning, Command Execution, Media Processing)         |
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|                                                                 |
|   Agent Protocol over MoQ Frame Layer (Header + Payload)        |
|                                                                 |
|  +-------------------+  +-------------------------------+      |
|  |   Frame Header    |  |       Frame Payload           |      |
|  | - Magic Number    |  | - CMD (JSON)                  |      |
|  | - Version         |  | - DATA (Binary chunks)        |      |
|  | - Message Type    |  | - STREAM (Media samples)      |      |
|  | - Session ID      |  | - HEARTBEAT (Empty)           |      |
|  | - Stream ID       |  |                               |      |
|  | - Flags           |  |                               |      |
|  | - Payload Length  |  |                               |      |
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|              Media Transport Layer (MoQ)                        |
|                                                                 |
|   Track Management  |  Group/Object  |  Datagram Support      |
|   Subscribe/Publish |  Prioritization|  Low-latency           |
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|                  Transport Layer (QUIC)                         |
|                                                                 |
|   UDP-based  |  0-RTT Handshake  |  Stream Multiplexing        |
|   TLS 1.3    |  Loss Recovery    |  Flow Control               |
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```

Protocol Stack of Agent Protocol over MoQ

Agent to Agent Communication Frame Structure

Field	Length (bytes)	Description	MoQ Mapping
Magic Number	2	Identifier: 0xA2A2	Detects agent-to-agent traffic
Version	1	Protocol version (e.g., 0x01)	Forward compatibility
Msg Type	1	Message type: CMD(0x01), DATA(0x02), STREAM(0x03), HEARTBEAT(0x04)	Route to handler
Session ID	8	Unique session identifier (UUIDv7 or hash)	Correlate multi-turn dialogues
Stream ID	4	Identifier for fragmented streams	Reassemble large data
Flags	1	Bitmask: compressed(0x01), encrypted(0x02), fragmented(0x04)	Processing hints
Payload Len	4	Length of the following payload	Parse boundary

Usage Examples

```
A.3.1.  A.3.1 Real-time Video Stream for Design Feedback

   *Step 1: Session Handshake with Authorization*

   Agent 1 (Smartphone)                    Agent 2 (Cloud)
           |                                          |
           |  CMD {action:"session_init",             |
           |       session_id:"abc-123",              |
           |       auth_token:"<JWT>"}                |
           |----------------------------------------->|
           |                                          |
           |  CMD {action:"session_ack",              |
           |       session_id:"abc-123",              |
           |       status:"approved"}                 |
           |<-----------------------------------------|

                 Figure 4: Session Establishment Sequence
```

```
Agent 1 (Subscriber)                    Agent 2 (Publisher)
         |                                                |
         |  CMD {action:"subscribe_request",              |
         |       session_id:"abc-123",                    |
         |       track_name:"agent2.video.h264",          |
         |       subscriber_id:"agent1"}                  |
         |------------------------------------------------>|
         |                                                |
         |  CMD {action:"track_ready",                    |
         |       session_id:"abc-123",                    |
         |       status:"approved",                       |
         |       track_url:"moq://track/agent2.video.h264"}|
         |<-----------------------------------------------|

               Figure 6: Subscription Request Sequence
```

Scenario

- Agent 1: Personal Assistant
- Agent 2: Professional Designer